

Questions

MathonGo

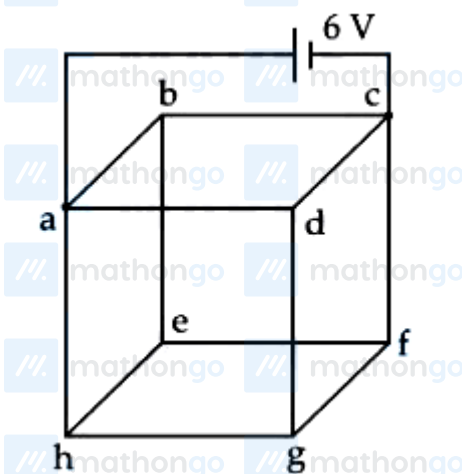
Q1 - 2024 (04 Apr Shift 1)

To measure the internal resistance of a battery, potentiometer is used. For $R = 10\Omega$, the balance point is observed at $l = 500$ cm and for $R = 1\Omega$ the balance point is observed at $l = 400$ cm. The internal resistance of the battery is approximately :

- (1) 0.2Ω
- (2) 0.3Ω
- (3) 0.4Ω
- (4) 0.1Ω

Q2 - 2024 (04 Apr Shift 1)

Twelve wires each having resistance 2Ω are joined to form a cube. A battery of 6 V emf is joined across point a and c . The voltage difference between e and f is _____ V.



Q3 - 2024 (04 Apr Shift 2)

An electric bulb rated 50 W – 200 V is connected across a 100 V supply. The power dissipation of the bulb is:

- (1) 25 W
- (2) 12.5 W
- (3) 50 W
- (4) 100 W

Questions

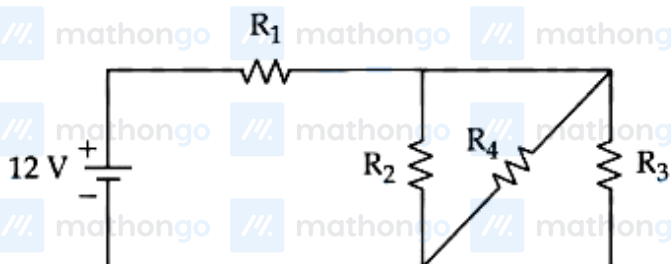
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Q4 - 2024 (04 Apr Shift 2)

Two wires A and B are made up of the same material and have the same mass. Wire A has radius of 2.0 mm and wire B has radius of 4.0 mm . The resistance of wire B is 2Ω . The resistance of wire A is _____ Ω .

Q5 - 2024 (05 Apr Shift 1)

In the given figure $R_1 = 10\Omega$, $R_2 = 8\Omega$, $R_3 = 4\Omega$ and $R_4 = 8\Omega$. Battery is ideal with emf 12 V . Equivalent resistant of the circuit and current supplied by battery are respectively :



(1) 10.5Ω and 1.14 A

(2) 12Ω and 1 A

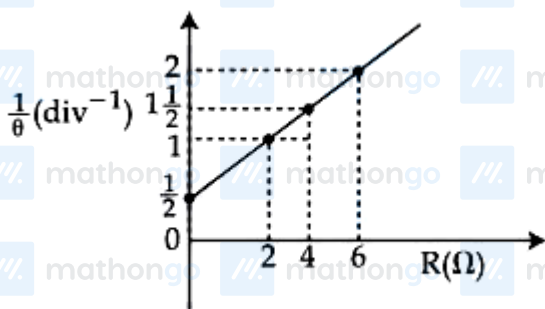
(3) 10.5Ω and 1 A

(4) 12Ω and 11.4 A

Q6 - 2024 (05 Apr Shift 1)

In the experiment to determine the galvanometer resistance by half-deflection method, the plot of $1/\theta$ vs the resistance (R) of the resistance box is shown in the figure. The figure of merit of the galvanometer is _____

$\times 10^{-1}\text{ A/division}$. [The source has emf 2V]



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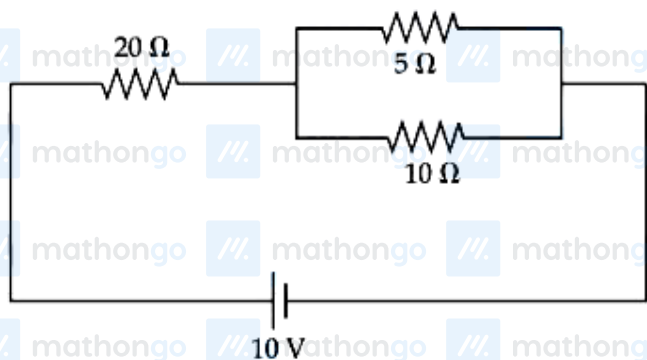
Q7 - 2024 (05 Apr Shift 2)

A galvanometer of resistance 100Ω when connected in series with 400Ω measures a voltage of upto 10 V. The value of resistance required to convert the galvanometer into ammeter to read upto 10 A is $x \times 10^{-2}\Omega$. The value of x is :

- (1) 2
- (2) 800
- (3) 20
- (4) 200

Q8 - 2024 (05 Apr Shift 2)

The ratio of heat dissipated per second through the resistance 5Ω and 10Ω in the circuit given below is :



- (1) 1 : 2
- (2) 2 : 1
- (3) 4 : 1
- (4) 1 : 1

Q9 - 2024 (05 Apr Shift 2)

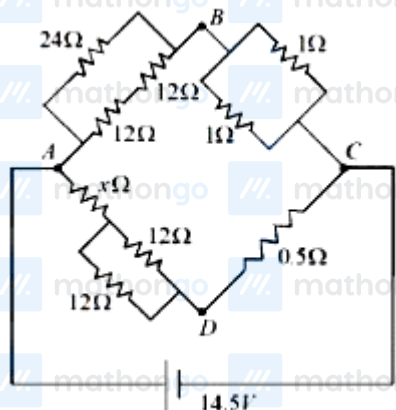
A wire of resistance 20Ω is divided into 10 equal parts, resulting pairs. A combination of two parts are connected in parallel and so on. Now resulting pairs of parallel combination are connected in series. The equivalent resistance of final combination is _____ Ω .

Questions

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Q10 - 2024 (06 Apr Shift 1)

The value of unknown resistance (x) for which the potential difference between B and D will be zero in the arrangement shown, is :



(1) 3Ω

(2) 42Ω

(3) 9Ω

(4) 6Ω

Q11 - 2024 (06 Apr Shift 1)

A wire of resistance R and radius r is stretched till its radius became $r/2$. If new resistance of the stretched wire is xR , then value of x is _____

Q12 - 2024 (06 Apr Shift 2)

The number of electrons flowing per second in the filament of a 110 W bulb operating at 220 V is : (Given

$$e = 1.6 \times 10^{-19} \text{ C})$$

(1) 6.25×10^{17}

(2) 1.25×10^{19}

(3) 6.25×10^{18}

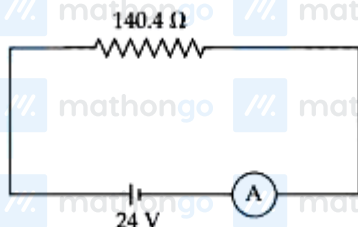
(4) 31.25×10^{17}

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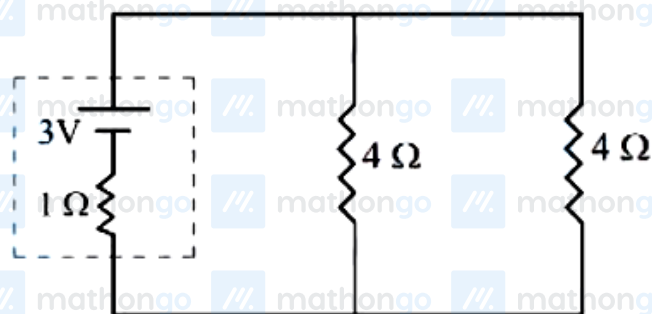
Q13 - 2024 (06 Apr Shift 2)

In the given figure an ammeter A consists of a 240Ω coil connected in parallel to a 10Ω shunt. The reading of the ammeter is _____ mA



Q14 - 2024 (08 Apr Shift 1)

In the given circuit, the terminal potential difference of the cell is :

(1) 2 V (2) 3 V (3) 4 V (4) 1.5 V

Q15 - 2024 (08 Apr Shift 1)

Resistance of a wire at 0°C , 100°C and $t^\circ\text{C}$ is found to be 10Ω , 10.2Ω and 10.95Ω respectively. The temperature t in Kelvin scale is _____

Q16 - 2024 (08 Apr Shift 2)

Questions

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Water boils in an electric kettle in 20 minutes after being switched on. Using the same main supply, the length of the heating element should be _____ to _____ times of its initial length if the water is to be boiled in 15 minutes.

(1) decreased, $3/4$

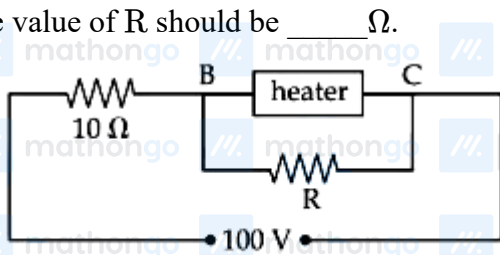
(2) increased, $4/3$

(3) decreased, $4/3$

(4) increased, $3/4$

Q17 - 2024 (08 Apr Shift 2)

A heater is designed to operate with a power of 1000 W in a 100 V line. It is connected in combination with a resistance of 10Ω and a resistance R , to a 100 V mains as shown in figure. For the heater to operate at 62.5 W, the value of R should be _____ Ω .



Q18 - 2024 (09 Apr Shift 1)

A galvanometer has a coil of resistance 200Ω with a full scale deflection at $20\mu\text{A}$. The value of resistance to be added to use it as an ammeter of range $(0 - 20)\text{mA}$ is ;

(1) 0.40Ω

(2) 0.20Ω

(3) 0.50Ω

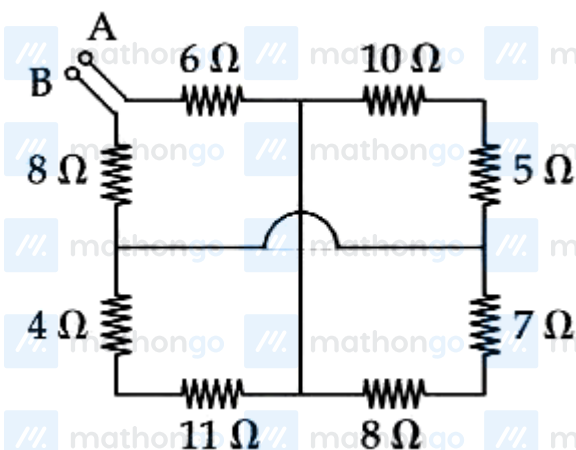
(4) 0.10Ω

Q19 - 2024 (09 Apr Shift 1)

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The equivalent resistance between A and B is :



(1) 18Ω

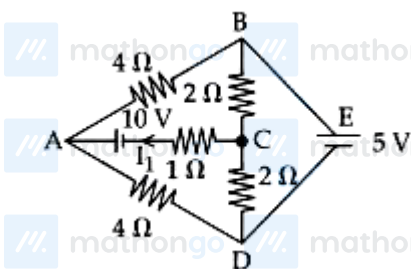
(2) 19Ω

(3) 25Ω

(4) 27Ω

Q20 - 2024 (09 Apr Shift 1)

The current flowing through the 1Ω resistor is $\frac{n}{10}$ A. The value of n is _____

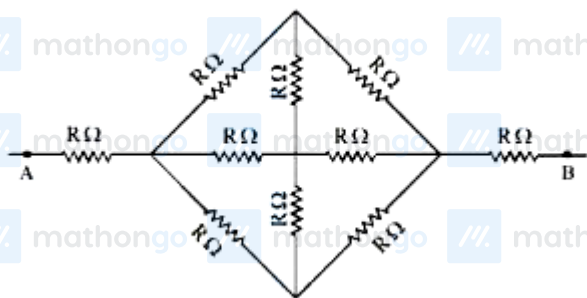


Q21 - 2024 (09 Apr Shift 2)

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The effective resistance between A and B , if resistance of each resistor is R , will be



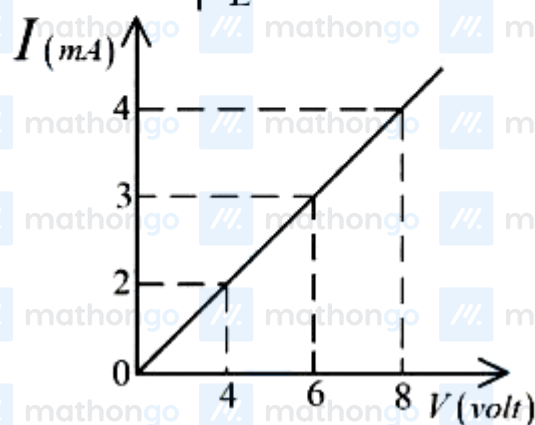
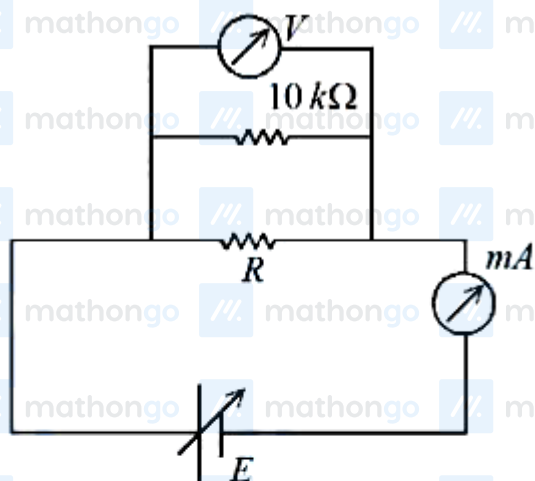
- (1) $\frac{8R}{3}$
 (2) $\frac{5R}{3}$
 (3) $\frac{4R}{3}$
 (4) $\frac{2}{3}R$

Q22 - 2024 (09 Apr Shift 2)

To determine the resistance (R) of a wire, a circuit is designed below. The $V - I$ characteristic curve for this circuit is plotted for the voltmeter and the ammeter readings as shown in figure. The value of R is _____ Ω .

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Answer Key

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Solutions

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Q1

Let potential gradient be λ .

$$\therefore i \times 10 = \lambda \times 500 = \varepsilon - i r_s$$

$$\Rightarrow 500\lambda = \varepsilon - 50\lambda r_s$$

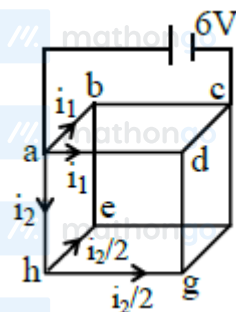
Also,

$$i' \times 1 = \lambda \times 400 = \varepsilon - i' r_s$$

$$\Rightarrow 400\lambda = \varepsilon - 400\lambda r_s$$

$$\therefore 100\lambda = 350\lambda r_s \Rightarrow r_s = \frac{10}{35} \approx 0.3\Omega$$

Q2



From symmetry, current through e-b & g-d = 0

$$\therefore R_{eq} = \frac{3}{4} \times R = \frac{3}{2}\Omega$$

$$\therefore \text{Current through battery} = \frac{6 \times 2}{3} = 4 \text{ A}$$

$$i_2 = \frac{4}{8} \times 2 = 1 \text{ A}$$

$$\therefore \Delta V \text{ across e-f} = \frac{i_2}{2} \times R = \frac{1}{2} \times 2 = 1 \text{ V}$$

Q3

Rated power & voltage gives resistance

$$R = \frac{V^2}{P} = \frac{(200)^2}{50} = \frac{40000}{50}$$

$$R = 800$$

$$P = \frac{(V_{\text{applied}})^2}{R} = \frac{(100)^2}{800}$$

$$P = 12.5 \text{ watt}$$

Q4

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Solutions

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$$\therefore R = \frac{\rho \ell}{A} = \frac{\rho V}{A^2}$$

$$\therefore \frac{R_A}{R_B} = \frac{A_B^2}{A_A^2} = \frac{r_B^4}{r_A^4}$$

$$\Rightarrow \frac{R_A}{2} = \left[\frac{4 \times 10^{-3}}{2 \times 10^{-3}} \right]^4$$

$$\Rightarrow R_A = 32\Omega.$$

Q5

Here R_2, R_3, R_4 are in parallel

$$\frac{1}{R_{234}} = \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$$

$$R_{234} = 2\Omega$$

 R_{234} is in series with R_1 so

$$R_{eq} = R_{234} + R_1 = 2 + 10 = 12\Omega$$

$$i = \frac{12}{12} = 1\text{Amp}$$

Q6

$$i = K\theta$$

$$\frac{2}{G+R} = K\theta$$

$$\Rightarrow \frac{1}{\theta} = \frac{(G+R)K}{2} = R\left(\frac{K}{2}\right) + \frac{KG}{2}$$

$$\text{Slope} = \frac{K}{2} = \frac{1}{4} \Rightarrow K = 0.5 = 5 \times 10^{-1} \text{A}$$

Q7

$$i_g = \frac{10}{400+100} = 20 \times 10^{-3} \text{A}$$

For ammeter

Let shunt resistance = S

$$i_g R = (i - i_g) S$$

$$20 \times 10^{-3} \times 100 = 10 S$$

$$S = 20 \times 10^{-2} \Omega$$

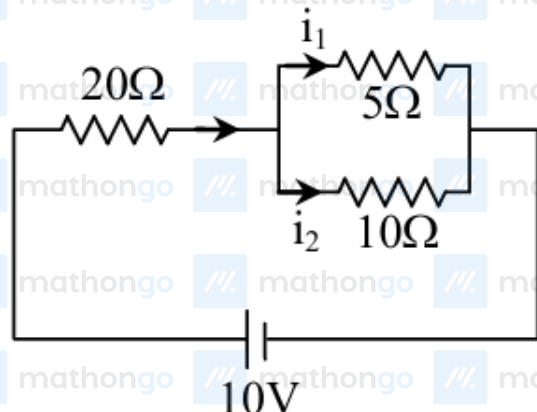
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Solutions

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$$\frac{i_1}{i_2} = \frac{10}{5} = \frac{2}{1}$$

$$\frac{P_1}{P_2} = \frac{i_1^2 R_1}{i_2^2 R_2} = \left(\frac{2}{1}\right)^2 \times \frac{5}{10} = \frac{2}{1}$$

Q9



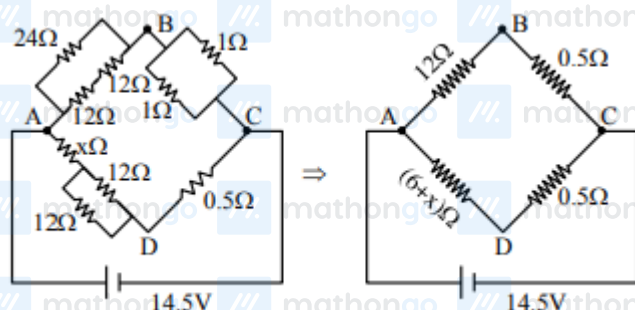
Each part has resistance = 2Ω

2 parts are connected in parallel so, $R = 1\Omega$

Now, there will be 5 parts each of resistance 1Ω , they are connected in series.

$$R_{eq} = 5R, R_{eq} = 5\Omega$$

Q10



In case of balanced Wheatstone Bridge

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Solutions

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$$\frac{V_{AB}}{V_{AD}} = \frac{V_{BC}}{V_{CD}} \Rightarrow \frac{12}{6+x} = \frac{0.5}{0.5}$$

$$x = 6\Omega$$

Q11

We know $R = \frac{\rho l}{A}$, $R \propto \frac{l}{r^2}$

As we stretch the wire, its length will increase but its radius will decrease keeping the volume constant

$$V_i = V_f$$

$$\pi r^2 l = \pi \frac{r_f^2}{4} l_f$$

$$l_f = 4l$$

$$\frac{R_{\text{new}}}{R_{\text{old}}} = \left(\frac{4l}{\frac{r^2}{4}} \right) \frac{r^2}{l} = 16$$

$$R_{\text{new}} = 16R$$

$$\therefore x = 16$$

Q12

Power (P) = V.I

$$\Rightarrow 110 = (220)(I)$$

$$\Rightarrow I = 0.5 \text{ A}$$

Now, $I = \frac{n \cdot e}{t}$

$$\Rightarrow 0.5 = \left(\frac{n}{t} \right) (1.6 \times 10^{-19})$$

$$\Rightarrow \frac{n}{t} = \frac{0.5}{1.6 \times 10^{-19}}$$

$$\Rightarrow \frac{n}{t} = 31.25 \times 10^{17}$$

Q13



Req

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Solutions

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$$= 140.4 + \frac{240 \times 10}{240 + 10}$$

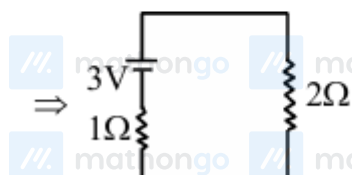
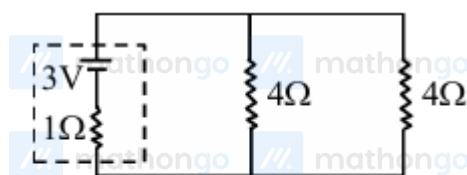
$$R_{eq} = 140.4 + \frac{2400}{250}$$

$$R_{eq} = 150\Omega$$

$$\therefore \text{Current in ammeter} = \frac{24}{150}$$

$$= 160 \text{ mA}$$

Q14



$$i = \frac{3}{1 + 2} = 1 \text{ A}$$

$$v = E - ir$$

$$= 3 - 1 \times 1 = 2 \text{ V}$$

Q15

$$R = R_0(1 + \alpha\Delta T)$$

$$\frac{\Delta R}{R_0} = \alpha\Delta T$$

Case-I

$$0^\circ\text{C} \rightarrow 100^\circ\text{C}$$

$$\frac{10.2 - 10}{10} = \alpha(100 - 0) \dots(1)$$

Case-II

$$0^\circ\text{C} \rightarrow t^\circ\text{C}$$

$$\frac{10.95 - 10}{10} = \alpha(t - 0) \dots(2)$$

$$\Rightarrow \frac{t}{100} = \frac{0.95}{0.2} = 475^\circ\text{C}$$

$$t = 475 + 273 = 748 \text{ K}$$

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Solutions

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Q16

$$P = \frac{v^2}{R}, R = \frac{\rho \ell}{A}$$

$$P \propto \frac{1}{\ell}$$

$$\frac{P_1}{P_2} = \frac{t_2}{t_1} = \frac{15}{20} = \frac{\ell_2}{\ell_1}$$

$$\ell_2 = \frac{3}{4} \ell_1$$

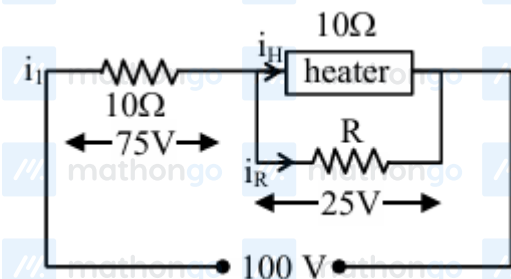
Q17

$$R_{\text{heater}} = \frac{V^2}{P} = \frac{(100)^2}{1000} = 10\Omega$$

$$\text{For heater } P = \frac{V^2}{R} \Rightarrow V = \sqrt{PR}$$

$$V = \sqrt{62.5 \times 10}$$

$$V = 25\text{V}$$



$$i_1 = \frac{75}{10} = 7.5 \text{ A}, i_H = \frac{25}{10} = 2.5 \text{ A}.$$

$$i_R = i_1 - i_H = 5$$

$$V = IR$$

$$R = \frac{25}{5} = 5\Omega$$

Q18

$$G = 200\Omega$$

$$i_g = 20\mu\text{A}$$

$$i = i_g \left(\frac{G}{S} + 1 \right)$$

$$\Rightarrow 20 \times 10^{-3} = 20 \times 10^{-6} \left(\frac{200}{S} + 1 \right)$$

$$\Rightarrow \frac{200}{S} = 999$$

$$\Rightarrow S \approx 0.2\Omega$$

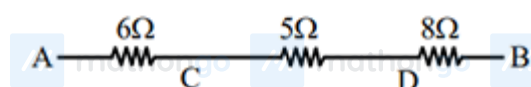
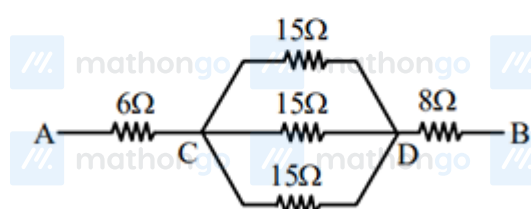
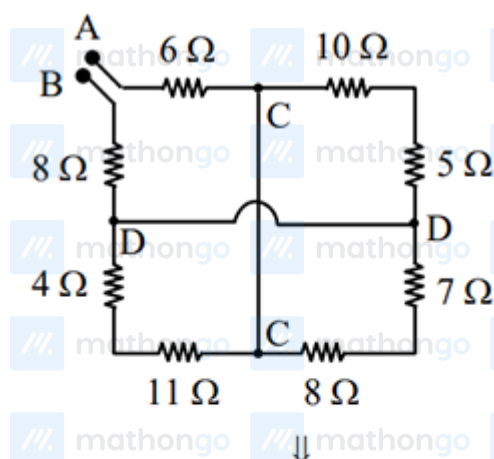
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Solutions

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Q19



$$\Rightarrow R_{\text{eq}} = 6\Omega + 5\Omega + 8\Omega = 19\Omega$$

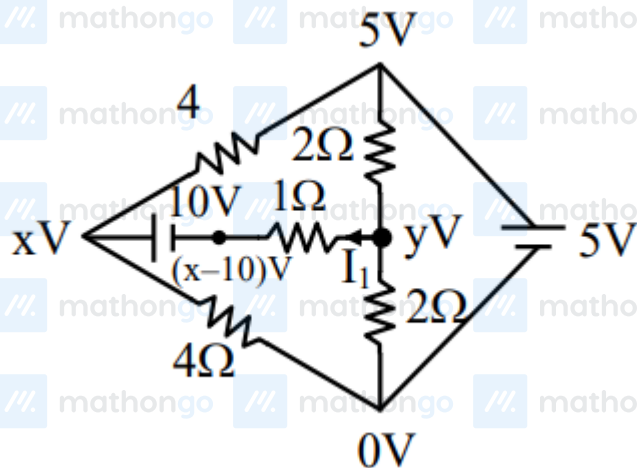
Q20

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$$\frac{y-5}{2} + \frac{y-0}{2} + \frac{y-x+10}{1} = 0$$

$$y-5+y+2y-2x+20=0$$

$$4y-2x+15=0 \quad \dots (i)$$

$$\frac{x-5}{4} + \frac{x-0}{4} + \frac{x-10-y}{1} = 0$$

$$x-5+x+4x-40-4y=0$$

$$6x-4y-45=0 \quad \dots (ii)$$

$$-2x+4y+15=0 \quad \dots (iii)$$

$$4x-30=0$$

$$x = \frac{15}{2} \quad \& 4y-15+15=0$$

$$y=0$$

$$i = \frac{y-x+10}{1}$$

$$i = \frac{0-7.5+10}{1}$$

$$i = 2.5A = \frac{n}{10}A$$

$$n = 25$$

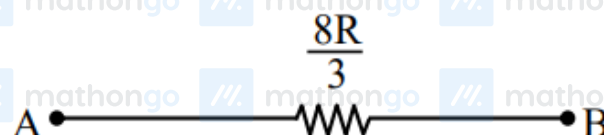
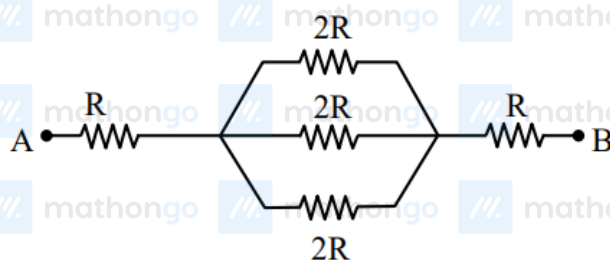
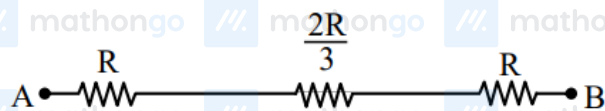
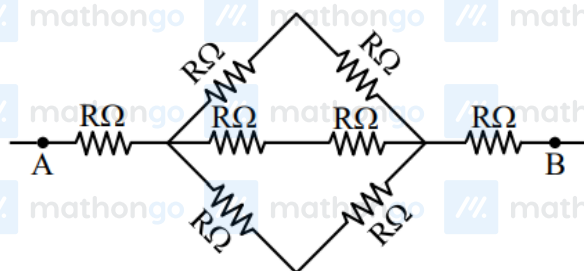
Q21

From symmetry we can remove two middle resistance.

New circuit is

Solutions

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Q22

$$R_{eq} = \frac{10^4 R}{10^4 + R}$$

$$E = 4V, I = 2mA$$

$$I = \frac{E}{R_{eq}} \Rightarrow 2 \times 10^{-3} = \frac{4(10^4 + R)}{10^4 R}$$

$$\Rightarrow 20R = 40000 + 4R$$

$$16R = 40000$$

$$R = 2500\Omega$$

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